

EUROPEAN PATENT APPLICATION EP 4 244 560 A1

FREEZE DRYER AND METHOD FOR FREEZE DRYING

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ABSTRACT

The invention relates to a freeze dryer, comprising a freeze-drying chamber having an inlet for allowing material to be dried to be introduced into the freeze-drying chamber and configured for being evacuated at least partially; and a vacuum source in fluid communication with the drying chamber and configured for evacuating the freeze-drying chamber at least partially, wherein a vacuum fluid path is defined between the freeze-drying chamber and the vacuum source, wherein the freeze dryer further comprises a material collector configured for collecting dried material from the freeze-drying chamber, which material collector is arranged outside the freeze-drying chamber and within the vacuum fluid path between the freeze-drying chamber and the vacuum source.

DESCRIPTION OF THE INVENTION

Background and Prior Art

Such a freeze-dryer is for example known from EP 1 601 919 B1 disclosing a method and device for freeze-drying solutions and liquid containing solid substances, in which a vessel is applied having a downwardly conical shape with inside a rotating mixing member that moves with a small interspace along the wall of the vessel. A disadvantage of the known freeze-dryer may be that the part of the material to be dried that is dried already remains within the freeze-drying chamber while drying the remaining of the material to be dried. Therefore, all product is continuously worked during the entire duration of the drying process through the action of the rotating mixing member. This is disadvantageous for applications where sensitivity to shear is of critical importance, such as probiotics, living cells, or spherical PLGA formulations. Another disadvantage may be that the available heat transfer area is used inefficiently, due to decreasing product volume as the freeze-drying progresses.

The Dynamic Elutriation Solution

According to the invention, the freeze dryer comprises a material collector arranged outside the freeze-drying chamber and within the vacuum fluid path between the freeze-drying chamber and the vacuum source. During use, as sublimation progresses, individual ice particles shrink in size, resulting in the release of a powdery dust of the material. The powdery dust flows towards the material collector along with the sublimated vapor flow, where it is deposited on a heated filter candle. By removing dried material from the chamber during active sublimation, the dried material is worked as short as possible, eliminating shear-induced cell rupture and particle abrasion. Furthermore, the remaining wet bed occupies the heat transfer wall with high efficiency.

Bypass Conduit (210) and Intermediate Isolation Valve (200)

In a preferred embodiment, a butterfly valve (200) is placed between the vessel outlet and the material collector, and a bypass conduit (210) bypasses the valve. During freezing and early sublimation, the valve (200) is OPEN, allowing large un-sublimated ice crystals to fall back into the conical vessel by gravity. When dry dust begins releasing, the valve (200) is CLOSED, forcing vapor and dry dust through the high-velocity bypass conduit (210) to the filter screen (176). The filtered powder falls onto the top of the closed valve (200), which temporarily holds the collected dry powder. Pulsed wall gas bleed nozzles (99) inject micro-bursts of nitrogen to strip boundary-layer dust and entrain it into the bypass conduit.

PATENT CLAIMS (Summary of Claims 1 to 15)

1. Freeze dryer, comprising: a freeze-drying chamber (3, 103) configured for being evacuated at least partially; a vacuum source in fluid communication with the drying chamber; and a material collector (40, 140) configured for collecting dried material from the freeze-drying chamber, arranged outside the freeze-drying chamber and within the vacuum fluid path between the chamber and the vacuum source.

2. Freeze dryer according to claim 1, wherein the material collector comprises a collector housing with a double-jacketed wall configured for receiving a heating fluid to heat fluid and/or collected dried material.
3. Freeze dryer according to claim 1 or 2, wherein the material collector comprises a collecting filter screen (76, 176) dividing the collector fluid path into upstream and downstream portions.
4. Freeze dryer according to any one of claims 1 to 3, further comprising a valve (200) arranged between the freeze-drying chamber and the material collector, movable between an open position and a closed position.
5. Freeze dryer according to claim 4, wherein the valve (200) is configured for temporarily holding material collected by the material collector when in the closed position.
6. Freeze dryer according to claim 4 or 5, further comprising a fluid bypass conduit (210) providing a fluid path between the freeze-drying chamber and the material collector which bypasses the valve (200).
7. Freeze dryer according to any one of the preceding claims, further comprising a purge inlet (90) configured to provide a reverse blow pulse to the material collector.
8. Freeze dryer according to any one of the preceding claims, wherein the freeze-drying chamber is defined within a substantially conical vessel (2, 102) with an agitating screw (16) orbiting along the vessel wall.
9. Freeze dryer according to claim 8, wherein the conical vessel comprises one or more blow nozzles (99) oriented into the chamber and connected to a gas source to generate an intermittent gas bleed.
10. Freeze dryer according to claim 8 or 9, wherein the conical vessel comprises a flush spherical segment valve (15) at the bottom apex thereof.